REQUEST FOR PROPOSAL

for

Digital Wallet

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December 19, 2025

Contents

1 Introduction

1.1 Purpose

This document specifies the functional and non-functional objectives for the development of a Digital Wallet System, a mobile financial application designed to enable secure, fast, and intuitive financial transactions via personal mobile devices. The system aims to meet the growing demand for digital alternatives to traditional payment methods based on physical cards and cash. This document follows IEEE Standard 29148-2018 for system and software requirements specification.

1.2 Scope

The **Digital Wallet System** is a mobile financial application (native iOS/Android) that provides secure payment and fund transfer capabilities through a mobile device interface.

Application:

- User account registration and PIN-based authentication with account lockout
- Peer-to-peer fund transfers with atomic transaction guarantees
- Contactless payments at terminals via QR Code and NFC technologies
- Transaction history and current balance display
- Automatic multi-currency conversion for international transactions
- Deposit and withdrawal functionalities linked to external banking systems

Benefits:

- Accessibility: Payments always available on mobile device
- Security: PIN authentication, account lockout
- Speed: Fast transactions ($<$ 2 seconds) with reliable processing
- Cost efficiency: Eliminates physical card costs

Objectives and Goals:

- Provide secure, user-friendly financial transactions on mobile devices

- Achieve 99.9% system uptime with atomic transaction processing
- Support multi-currency transactions with automatic conversion
- Maintain complete financial integrity

1.3 Product Overview

1.3.1 Product Perspective

The Digital Wallet operates as a standalone financial platform integrated with external banking systems and payment terminals. It functions as an intermediary between users and payment infrastructure, providing a unified interface for all payment-related operations.

1.3.2 Product Functions

The Digital Wallet System performs ten major functional areas:

- **User Registration:** User account creation with email, phone, and personal information validation
- **Authentication:** PIN-based identity verification with account lockout protection after 3 failed attempts
- **PIN Management:** PIN setup during registration and PIN change for security
- **Fund Transfer:** Peer-to-peer transfers between registered users with atomic transaction processing
- **Terminal Payments:** Contactless payments via QR Code and NFC at merchant terminals
- **Transaction History:** View, filter, search, and export transaction records
- **Balance Inquiry:** Check current account balance and available funds
- **Deposit Funds:** Add funds to wallet account from external banking systems
- **Withdraw Funds:** Withdraw funds from wallet to external bank accounts
- **Automatic Interest Application:** Apply accrued interest to user accounts on scheduled basis

Detailed functional specifications for each use case are provided in Section 3.2 (UC-01 through UC-10). Associated user-level scenarios demonstrating system behavior are included in Sections 3.3 (Usability) and 3.4 (Performance).

1.3.3 User Characteristics

Target User Groups

The Digital Wallet System is designed for the following user categories:

- **Age Range:** The system is primarily intended for users aged 18 to 65, representing approximately 85% of the expected user base. However, the system should support users outside this age range, including seniors with reduced mobility and limited digital literacy.
- **Geographic Distribution:** Primarily European users (EUR-based transactions) with international extension support
- **Technology Proficiency:** From novice (first-time mobile app users) to expert (power users with advanced app experience)
- **Financial Background:** Users with varying financial literacy levels (students, professionals, business owners, retirees)

Educational Level and Experience

Users possess diverse educational backgrounds (secondary education through university degree) and varying mobile app experience. Some users may be adopting smartphone technology for the first time, while others are frequent mobile banking users. The system must be usable by all groups without extensive training.

Technical Expertise

Technology familiarity ranges from basic (e.g., making calls, sending messages) to advanced (e.g., managing multiple apps, understanding security concepts). The system design must not assume technical knowledge and should guide users through complex operations via progressive disclosure and contextual help.

Accessibility and Physical Considerations

The system must accommodate:

- **Vision:** Users with color blindness, low vision; support text scaling and high-contrast modes
- **Hearing:** Users with hearing impairment; avoid reliance on audio-only feedback
- **Cognitive:** Users with attention difficulties; use simple language and clear navigation

Design Implications

These user characteristics drive the following design requirements:

- **Simplified UI:** Minimal cognitive load with clear visual hierarchy
- **Accessibility Compliance:** WCAG 2.1 Level AA minimum for color contrast, font sizes, and touch targets
- **Contextual Help:** In-app tooltips and help text for users unfamiliar with financial terminology
- **Multi-language Support:** Support for at least 15 languages to accommodate diverse user base

1.3.4 Limitations

Regulatory Requirements

- **PCI DSS 4.0 Compliance:** All payment processing must comply with Payment Card Industry Data Security Standards
- **GDPR Compliance:** User data collection, processing, and retention must comply with General Data Protection Regulation
- **ISO 27001:** Information security management system requirements must be met

Hardware and Platform Constraints

- **Supported Platforms:** iOS 13+ and Android 8+ only; older versions not supported
- **Network Connectivity:** System requires active internet connection for all financial transactions; offline mode limited to balance display
- **NFC/QR Capability:** Terminal payments require device NFC or camera support (not all devices have both)
- **Touch Screen:** Touch-based interface requires capacitive touchscreen; stylus support not required

System Interfaces and External Dependencies

- **Banking System Integration:** Dependent on third-party banking API availability and uptime; system cannot exceed banking backend performance
- **Payment Terminal Compatibility:** Only compatible with QR Code and NFC-enabled terminals; legacy terminal support not provided

- **Currency Exchange Data:** Real-time exchange rates sourced from external provider; conversion accuracy limited to provider precision
- **SMS/Email Services:** Verification codes and notifications dependent on third-party SMS and email service providers

Data and Quality Requirements

- **Transaction Limits:** Minimum 0.01 EUR, maximum 10,000 EUR per transaction; daily user limit 50,000 EUR (sum of all transactions carried out in a 24-hour period).

Criticality and Safety

- **Financial Criticality:** System handles real money; transaction failures are security incidents requiring immediate investigation and user notification
- **Data Integrity:** No data corruption or loss tolerance; atomic transactions are mandatory
- **User Credentials:** PIN must be changed at least annually

1.4 Definitions

- **Account Lockout:** Security mechanism that prevents access to an account after a specified number of failed authentication attempts.
- **Atomic Transaction:** Operation guaranteed to either complete in full or be completely reversed.
- **BDD (Behavior Driven Development):** Development methodology using 'behave' framework for scenario specification.
- **Circuit Breaker Pattern:** Design pattern used to detect failures and prevent the application from trying to perform an action that is doomed to fail.
- **Fitts's Law:** Predictive model of human movement stating that the time to acquire a target is a function of the distance to and size of the target.
- **Fixed-Point Arithmetic:** Decimal values represented as integers (e.g., cents, not euros) to prevent floating-point errors.
- **Immutable Audit Log:** Permanent record of all system actions that cannot be altered or deleted.
- **Lazy Loading:** Design pattern that defers the initialization of an object or data retrieval until the point at which it is needed.

- **Load Balancing:** The process of distributing network traffic across multiple servers to ensure no single server bears too much demand.
- **NFC (Near Field Communication):** Wireless technology for contactless payment transmission.
- **Peer-to-Peer (P2P):** Decentralized interactions between two parties (users) without a third-party intermediary for the transfer logic itself.
- **PIN (Personal Identification Number):** 4-6 digit security code used for user authentication.
- **Privacy by Design:** Approach to systems engineering which takes privacy into account throughout the whole engineering process.
- **Progressive Disclosure:** Interaction design pattern that sequences information and actions across several screens to reduce cognitive load.
- **QR Code:** Matrix barcode used for payment initiation.
- **Sandboxing:** Security mechanism for separating running programs to mitigate system failures or software vulnerabilities.
- **Skeleton Screen:** UI pattern that shows a blank version of a page into which information is gradually loaded to improve perceived performance.
- **TPA (Third-Party Authority):** External payment terminals and POS systems.
- **Zero Sum Principle:** Total system money = sum of all user balances + system reserves (always true).

2 References

1. IEEE Std. 29148-2018, “Systems and software engineering – Life cycle processes – Requirements engineering,” IEEE, 2018.

2. ISO/IEC 27001:2013, “Information technology – Security techniques – Information security management systems – Requirements,” International Organization for Standardization, 2013.

3. PCI Security Standards Council, “Payment Card Industry Data Security Standard (PCI DSS), Version 4.0,” PCI Security Standards Council, Oct. 2022.

4. W3C, “Web Content Accessibility Guidelines (WCAG) 2.1,” World Wide Web Consortium, Jun. 2018. [Online]. Available: https://www.w3.org/WAI/WCAG21/quickref/

5. European Union, “Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data,” Official Journal of the European Union, May 2016.

6. IEEE Std. 754-2019, “IEEE Standard for Floating-Point Arithmetic,” IEEE, 2019.

3 Specific Requirements

3.1 External Interfaces

3.1.1 Use Case Diagram

Overview:

The use case diagram illustrates the functional interactions between actors and the Digital Wallet System. Each oval represents a use case (functional capability), and lines show relationships and dependencies.

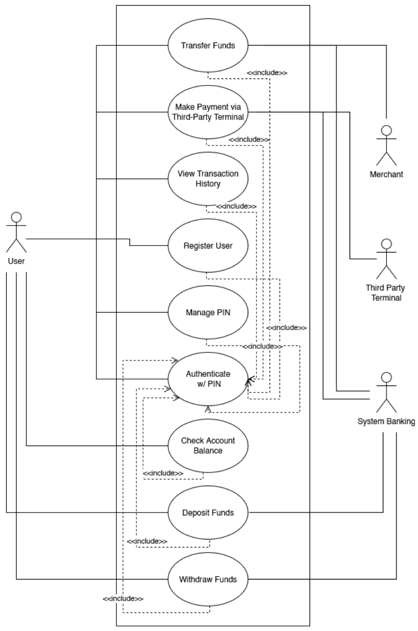

Figure 3.1: Use Case Diagram

Primary Actor:

The User (left side) is the main actor who initiates all wallet operations including registration, authentication, transfers, payments, and account management.

Secondary Actors:

- **Payment Terminal:** External POS/terminal system for contactless payments via NFC or QR Code
- **Banking System:** Third-party banking infrastructure that validates transactions, processes debits/credits, and manages account synchronization
- **Merchant:** Recipient of funds in terminal payment transactions

Use Case Summary:

- PIN authentication (UC-01)
- Fund transfers (UC-02)
- Terminal payments (UC-03)
- Transaction history (UC-04)
- PIN management (UC-05)
- User registration (UC-06)
- Balance inquiry (UC-07)
- Deposits (UC-08)
- Withdrawals (UC-09)
- Automatic Interest Application (UC-10).

This includes relationships ("<<include>>") that indicate mandatory dependencies.

3.1.2 Sequence Model Flow

Overview: The sequence diagram depicts the temporal interaction flow between actors and the system during a typical financial transaction.

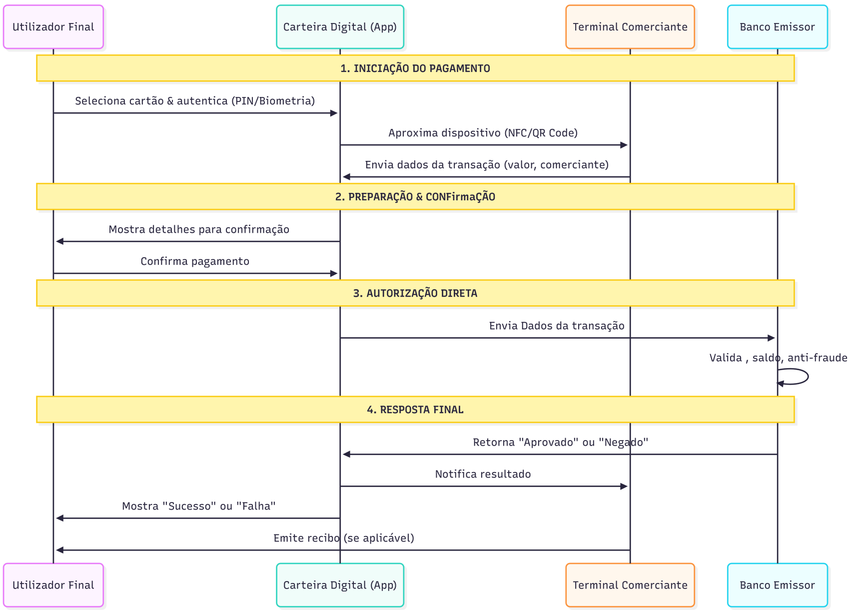

Figure 3.2: Sequence Model

Actors and Their Roles:

- **User:** Initiates transaction requests (e.g., enter PIN, select recipient, confirm payment)
- **Digital Wallet System:** Processes requests, validates data, manages balances, and communicates with external systems
- **Banking System:** Validates accounts, processes debits/credits, and returns transaction confirmations

Flow Stages:

1. **Input:** User provides credentials (PIN) and transaction details (recipient, amount, currency)
2. **Processing:** System validates PIN, checks balance, verifies recipient, converts currency if needed
3. **Output:** System generates transaction confirmation, updates balances, and notifies all parties

3.2 Functions

This section describes all functional requirements through use cases. Each use case details the interaction between users and the system, including preconditions, postconditions, and primary/alternate flows.

3.2.1 UC-01: Authenticate via PIN

- **Use Case ID:** UC-01
- **Use Case Name:** Authenticate via PIN
- **Primary Actor:** User
- **System:** Digital Wallet Application
- **Purpose:** Validate user identity and establish secure session
- **Precondition:** User has registered account with valid PIN
- **Postcondition Success:** User session authenticated and active
- **Postcondition Failure:** Authentication attempt logged; account locked after 3 failures
- **Main Flow:**

User	System
1. Launches Digital Wallet application	
	2. Displays PIN entry interface
3. Enters 4-6 digit PIN	
	4. Verifies account is not locked
	5. Validates PIN
	6. Creates session token valid for 30 minutes
	7. Displays authenticated dashboard

- **Alternate Flow**

4.1 - Account is Locked
Condition: At step 4, if the account is Locked

User	System
	5. Increments failed attempt counter
	6. If counter = 3, locks account for 24 hours
	7. Displays error: "Invalid PIN. Attempts remaining: {remaining}"
	8. If account locked, displays: "Account temporarily locked. Try again in 24 hours"

5.1 - Invalid PIN
Condition: At step 5, if PIN is incorrect

User	System
	5. Increments failed attempt counter
	6. If counter = 3, locks account for 24 hours
	7. Displays an error

3.2.2 UC-02: Transfer Funds

- **Use Case ID:** UC-02
- **Use Case Name:** Transfer Funds
- **Primary Actor:** User
- **Secondary Actor:** Banking System
- **System:** Digital Wallet Application
- **Purpose:** Enable user to transfer funds to another registered user
- **Precondition:** User authenticated; sufficient balance; recipient exists
- **Postcondition Success:** Funds transferred atomically; both accounts updated; receipt generated
- **Postcondition Failure:** No account changes; error message displayed
- **Main Flow:**

User	System	Banking System
1. Selects "Transfer Funds" from menu		
	2. Displays transfer form	
3. Enters recipient identifier (email or account number)		
4. Enters transfer amount		
5. Confirms transfer		
	6. Validates recipient exists	
	7. Validates sender balance $\geq$ transfer amount + fees	
	8. Initiates transaction	
		9. Confirms Transaction
		10. Sends Confirmation
	11. Displays confirmation with receipt number and timestamp	
	12. Sends notifications to both parties	

- **Alternate Flow**

7.1 - Insufficient Balance
Condition: At step 7, if balance insufficient

User	System
	8. Displays error: “Insufficient balance. Available: {amount} EUR”
	9. Returns to transfer form

6.1 - Recipient Not Found
Condition: At step 6, if recipient not found

User	System
	7. Displays error: “Recipient not found. Please verify account details”
	8. Returns to transfer form

5.1 - User Cancels Transfer
Condition: At step 5, if recipient not found

User	System
6. Selected the ”Undo” button	
	7. Temporarily discard the entered data.
	8. Displays a confirmation message: ”Operation canceled. No transfer was completed.”
	9. Return to the transfer screen (step 2 of the main flow)

10.1 - Banking System Rejects Transaction
Condition: At step 10, if banking system rejects the transaction

User	System	Banking System
		11. Transaction validation fails
		12. Sends error response with rejection code
	13. Receives rejection response	
	14. Displays an error	
	15. Refunds	

3.2.3 UC-03: Make Payment via Third-Party Terminal

- **Use Case ID:** UC-03
- **Use Case Name:** Make Payment at TPA Terminal
- **Primary Actor:** User
- **Secondary Actor:** Payment Terminal (TPA), Banking System
- **System:** Digital Wallet Application
- **Purpose:** Enable contactless payment at third-party terminals
- **Precondition:** User authenticated; NFC/QR capability enabled; sufficient balance
- **Postcondition Success:** Payment processed; terminal notified; receipt issued
- **Postcondition Failure:** Transaction rolled back; error displayed
- **Main Flow - QR Code Payment:**

User	System	Terminal	Banking System
1. User approaches payment terminal			
		2. Displays QR code containing: merchant ID, amount, currency, timestamp	
3. User opens wallet and selects “Scan QR Code”			
	4. Camera scans QR code		
	5. Displays merchant name and payment amount for confirmation		
6. User confirms payment			
	7. Validates amount		
	8. Send Transaction Information		
			9. Processes the Information
			10. Sends Response
	11. Receives Response		
	12. Sends confirmation to terminal		
		13. Displays success and issues receipt	
	14. Displays confirmation to user		

- **Main Flow - NFC Payment:**

User	System	Terminal	Banking System
1. User approaches NFC-enabled terminal			
		2. Transmits payment request via NFC	
	3. Displays amount and merchant for confirmation		
4. User confirms payment			
	5. Validates amount		
	6. Validates merchant		
	7. Send Transaction Information		
			8. Processes the Information
			9. Sends Response
	10. Sends confirmation to terminal		
		11. Displays success and issues receipt	
	12. Displays confirmation to user		

- **Alternate Flow**

5.1 - User Cancels Payment
Condition: At step 5, User identifies an error or decides to cancel the payment.

User	System	Terminal
	6. Displays a 'Back' button	
7. Clicks on the 'Back' button		
	8. System cancels the payment without charging	
	9. Returns to payment initiation	

9.1/10.1 - Banking System Rejects Payment (QR Code and NFC)

Condition: At step 10 (QR Code flow) and at step 9 (NFC Flow), banking system rejects the transaction

User	**System**	**Terminal**	**Banking System**
			Transaction validation fails
			Sends rejection error to system
	Receives rejection error		
	Sends error notification to terminal		
		Displays an error	
	Displays an error to user		
	No funds deducted		

3.2.4 UC-04: View Transaction History

- **Use Case ID:** UC-04
- **Use Case Name:** View Transaction History
- **Primary Actor:** User
- **System:** Digital Wallet Application
- **Purpose:** Enable user to review past transactions and account activity
- **Precondition:** User authenticated
- **Postcondition:** Transaction history displayed with filtering and export options
- **Main Flow:**

User	**System**
1. Selects "Transaction History" from menu	
	2. Retrieves all transactions for authenticated user
	3. Displays transactions in reverse chronological order
	4. Displays details: date, time, counterparty, amount, type, status

3.2.5 UC-05: Manage PIN

- **Use Case ID:** UC-05
- **Use Case Name:** Manage PIN Settings
- **Primary Actor:** User
- **System:** Digital Wallet Application
- **Purpose:** Enable user to change PIN and manage authentication
- **Precondition:** User authenticated
- **Postcondition:** PIN changed or reset successfully
- **Main Flow - Change PIN:**

User	System
1. Navigates to "Security Settings"	
	2. Displays security options menu
3. Selects "Change PIN"	
	4. Prompts for current PIN (re-authentication)
5. Enters current PIN	
	6. Validates current PIN
	7. Prompts for new PIN (4-6 digits)
8. Enters new PIN twice for confirmation	
	9. Validates PIN meets complexity requirements
	10. Updates PIN
	11. Displays success

3.2.6 UC-06: Register User

- **Use Case ID:** UC-06
- **Use Case Name:** Register User
- **Primary Actor:** User
- **System:** Digital Wallet Application
- **Purpose:** Create a new user account and set up security credentials
- **Precondition:** Application installed on device; device has internet connectivity
- **Postcondition Success:** User account created; PIN set; user authenticated
- **Postcondition Failure:** No account created; error message displayed
- **Main Flow:**

User	System
1. Launches application and selects "Register"	
	2. Displays registration form (Email, Phone, Name)
3. Enters requested personal information	
	4. Validates input format and uniqueness
	5. Sends verification code to phone
	6. Prompts for verification code
7. Enters received verification code	
	8. Validates code
	9. Prompts for PIN creation (4-6 digits)
10. Enters new PIN twice	
	11. Validates PIN complexity
	12. Creates user account and wallet
	13. Logs user in and displays dashboard

- **Alternate Flow**

4.1 - User Already Exists
Condition: At step 4, if email or phone is already registered

User	System
	5. Displays error: "Account already exists"
	6. Offers option to recover account or login

8.1 - Code Incorrect
Condition: At step 8, if verification code is incorrect

User	**System**
	9. Code not valid
	10. Display Error
	11. Send another code

3.2.7 UC-07: Check Account Balance

- **Use Case ID:** UC-07
- **Use Case Name:** Check Account Balance
- **Primary Actor:** User
- **System:** Digital Wallet Application
- **Purpose:** Allow user to view their current available funds
- **Precondition:** User authenticated
- **Postcondition:** Current balance displayed accurately
- **Main Flow:**

User	System
1. Accesses the Dashboard (Home Screen)	
	2. Retrieves current balance
	3. Displays balance in large, legible format

3.2.8 UC-08: Deposit Funds

- **Use Case ID:** UC-08
- **Use Case Name:** Deposit Funds (Top-up)
- **Primary Actor:** User
- **Secondary Actor:** Banking System
- **System:** Digital Wallet Application
- **Purpose:** Load money into the wallet from an external bank account
- **Precondition:** User authenticated; linked bank account or card available
- **Postcondition Success:** Wallet balance increased; Bank account debited
- **Postcondition Failure:** Balance unchanged; error message displayed
- **Main Flow:**

User	System	Banking System
1. Selects “Add Money” / “Deposit”		
	2. Prompts for amount and source method	
3. Enters amount and selects Bank Account		
4. Confirms transaction		
	5. Validates amount limits	
	6. Sends debit request	
		7. Processes debit request
		8. Sends Confirmation
	9. Credits user wallet balance (Atomic)	
	10. Displays success message	

- **Alternate Flow**

8.1 - Banking System Rejects Debit Request
Condition: At step 8, if the banking system rejects the debit request

User	System	Banking System
		8. Transaction validation fails
		9. Sends rejection error with code
	10. Receives rejection response	
	12. Rolls back any wallet balance changes	
	13. Displays an error	

3.2.9 UC-09: Withdraw Funds

- **Use Case ID:** UC-09
- **Use Case Name:** Withdraw Funds
- **Primary Actor:** User
- **Secondary Actor:** Banking System
- **System:** Digital Wallet Application
- **Purpose:** Move money from wallet back to external bank account
- **Precondition:** User authenticated; sufficient wallet balance
- **Postcondition Success:** Wallet balance decreased; Bank account credited
- **Main Flow:**

User	System	Banking System
1. Selects "Withdraw"		
	2. Displays linked bank accounts	
3. Selects destination IBAN and amount		
4. Confirms withdrawal		
	5. Validates wallet balance $\geq$ amount	
	6. Debits wallet balance (Atomic)	
	7. Sends credit request	
		8. Processes credit to user account
		9. Sends Confirmation
	10. Records transaction status as Completed	
	11. Notifies user	

- **Alternate Flow**

9.1 - Banking System Rejects Withdrawal
Condition: At step 9, if banking system rejects the credit to user's bank account

User	System	Banking System
		10. Account credit validation fails
		11. Sends rejection error with code
	12. Receives rejection response	
	13. Re-credits wallet balance	
	14. Displays an error	

3.2.10 UC-10: Apply Interest

- **Use Case ID:** UC-10
- **Use Case Name:** Apply Interest on Balances
- **Primary Actor:** System (Time Trigger)
- **System:** Digital Wallet Backend
- **Purpose:** Automatically calculate and credit interest to eligible user accounts
- **Precondition:** Interest period (e.g., end of month) reached; positive user balances
- **Postcondition:** Balances updated with interest; "Zero Sum" maintained via System Reserve
- **Main Flow:**

Trigger	**System**
1. Scheduled time reached	
	2. Retrieves all active accounts with positive balance
	3. Calculates interest
	4. Debits System Reserve Account (Source of interest)
	5. Credits User Accounts (Atomic batch operation)

3.3 Usability Requirements

3.3.1 Scenario 1: First-Time User Makes a Money Transfer

Quality Attribute: Usability
Source of Stimulus: A new user who just downloaded the Digital Wallet app
Stimulus: User wants to send money to a friend for the first time
Environment: Normal app usage, user at home, connected to the internet
Artifact: Digital Wallet mobile application
Response: The system guides the user step-by-step through the transfer process with clear instructions and helpful suggestions
Response Measure: Users rate the transfer process as "easy" or "very easy" in at least 85% of cases

Explanation:
The system design prioritizes intuitive interaction for first-time users to minimize cognitive load. The transfer interface must streamline the input process by offering contact suggestions and clearly labeling required fields. The confirmation step serves as a critical error-prevention mechanism, ensuring user confidence and data accuracy before transaction commitment. Post-transaction feedback confirms the operation's success, reinforcing system reliability.

3.3.2 Scenario 2: User Checks Account Balance During Busy Shopping

Quality Attribute: Usability
Source of Stimulus: A customer shopping in a busy store at peak hours
Stimulus: User quickly needs to check if they have enough money before making a payment
Environment: In-store shopping environment, touchscreen may be difficult, poor lighting
Artifact: Digital Wallet mobile application
Response: The system displays the account balance instantly in large, clear numbers that are easy to read even in bright sunlight **Response Measure:** Users successfully identify their available funds correctly 99% of the time
Explanation:
To support usability in high-distraction environments, the system must present critical financial information immediately upon authentication. The landing page layout prioritizes the account balance using high-contrast, large typography to ensure readability under varying lighting conditions. Interface elements must adhere to Fitts's Law, providing sufficiently large touch targets to prevent accidental inputs during mobile usage. Persistent user preferences ensure a consistent and efficient experience across sessions.

3.3.3 Scenario 3: User Recovers from a Mistake During Payment

Quality Attribute: Usability
Source of Stimulus: A user who made an error while entering payment details
Stimulus: User entered wrong recipient email or wrong amount and wants to cancel
Environment: Normal app usage, user realizes mistake before final confirmation
Artifact: Digital Wallet mobile application
Response: The system allows the user to easily undo or correct the mistake without losing their work or encountering confusing error messages
Response Measure: Users can cancel or go back to fix a mistake in less than 10 seconds

Explanation:

The system implements a robust error recovery mechanism by enforcing a confirmation step before transaction finalization. This intermediate state allows users to verify transaction details (recipient, amount, currency) against their intent. Navigation controls must enable non-destructive backtracking, preserving previously entered data to facilitate quick corrections without requiring re-entry of valid information, thereby reducing user frustration and transaction abandonment rates.

3.4 Performance Requirements

3.4.1 Scenario 1: Quick Payment During Busy Shopping Season

Quality Attribute: Efficiency (Speed and Responsiveness)
Source of Stimulus: A customer trying to pay at a store during the busy Christmas shopping season
Stimulus: User opens the wallet app and initiates a payment at the payment terminal
Environment: Peak shopping hours (evening, holiday season), thousands of other customers also shopping simultaneously
Artifact: Digital Wallet mobile application
Response: The system processes the payment quickly so the user can complete their purchase without long waiting times
Response Measure: The system processes the payment within a maximum of 2 seconds, measured from the moment the user confirms the transaction until the success message is displayed on the interface

Explanation:

System responsiveness is critical during peak load periods to prevent service bottlenecks at point-of-sale. The application is required to retrieve terminal information and process the transaction confirmation within the specified 2-second latency window, ensuring a seamless checkout experience regardless of concurrent system load. This performance benchmark must be maintained to ensure user trust and adoption in time-sensitive retail environments.

3.4.2 Scenario 2: Accessing Transaction History Over Slow Internet Connection

Quality Attribute: Efficiency (Responsiveness with Limited Resources)
Source of Stimulus: A user traveling in an area with slow or unstable internet connection
Stimulus: User opens the app and requests to view their transaction history from the past years
Environment: Poor network conditions (slow 3G or unstable mobile hotspot), user is on a bus or train
Artifact: Digital Wallet mobile application
Response: The system retrieves and displays the transaction history efficiently, even when dealing with large amounts of data and slow networks
Response Measure: Transaction history loads in less than 3 seconds on slow connections; 95% of users report the app feels responsive

Explanation:

The system must employ efficient data retrieval strategies, such as pagination and lazy loading, to handle large datasets under constrained network conditions. When accessing transaction history, the application prioritizes the display of the most recent records to provide immediate utility. Background fetching and caching mechanisms are utilized to ensure that subsequent data retrieval does not block the user interface, maintaining perceived performance even on high-latency networks.

3.4.3 Scenario 3: Rapid Consecutive Payments During Special Event

Quality Attribute: Efficiency (Sustained Performance Under Load)
Source of Stimulus: Multiple users making rapid, repeated payments at an event
Stimulus: Thousands of event attendees making quick payments within a short time window
Environment: Special event (concert, festival, conference), concentrated group of people, short time period (30 minutes)
Artifact: Digital Wallet mobile application
Response: The system maintains consistent performance and processes all payment requests fairly, without discrimination or delays
Response Measure: System handles 1,000 payments per second; each payment completes within 2 seconds; zero payments are lost or rejected due to overload

Explanation:

The backend architecture must support high concurrency to handle burst traffic typical of large-scale events. The system is designed to process a minimum of 1,000 transactions per second without degradation in response time or data integrity. Load balancing and scalable infrastructure ensure that simultaneous requests are queued and processed fairly, preventing service denial and ensuring that all valid transactions are completed within the defined SLA limits.

3.5 Logical Database Requirements

3.5.1 Data Entity Model (Entity-Relationship Diagram)

Entities

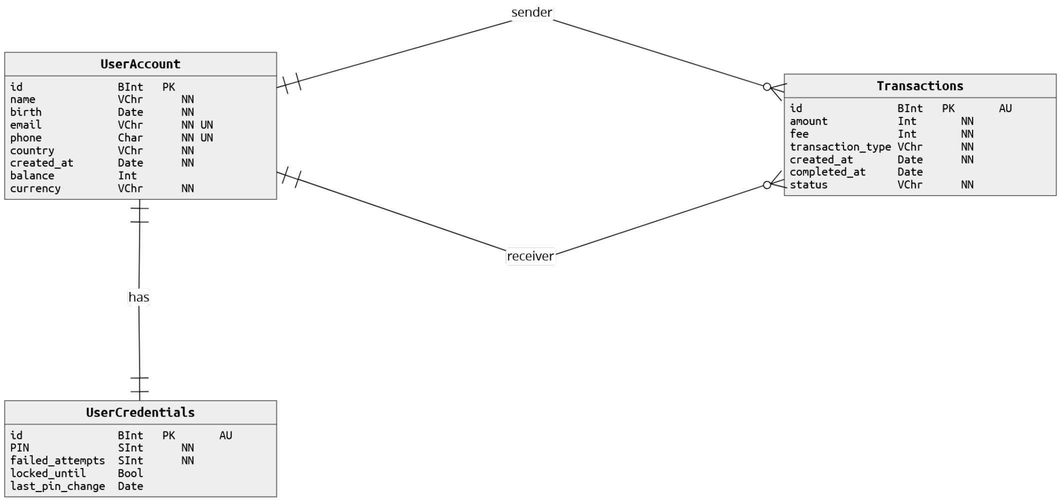

Figure 3.3: Entity Diagram

- **Entity: User**
  - **UserID** (Primary Key): UUID, immutable
  - **Name**: String, user-provided
  - **Birth**: Date, unique
  - **Email**: String, unique
  - **Phone**: String, unique
  - **Country**: ISO 3166-1 code
  - **CreatedAt**: Timestamp, immutable
  - **Balance**: Integer, non-negative, default=0
  - **Currency**: String, max length=3
- **Entity: User Credentials**
  - **UserID** (Primary Key): UUID, immutable
  - **PIN**: String
  - **FailedAttempts**: Integer, default=0
  - **LockedUntil**: Timestamp, NULL if not locked

  - **LastPINChange**: Timestamp

- **Entity: Transaction**
  - **ID** (Primary Key): UUID, immutable
  - **Amount**: Integer
  - **Fee**: Integer
  - **TransactionType**: String (TRANSFER, PAYMENT, DEPOSIT, WITHDRAWAL, FEE)
  - **CreatedAt**: Timestamp
  - **CompletedAt**: Timestamp, NULL until completed
  - **Status**: String (PENDING, COMPLETED, FAILED, REVERSED)

Relationships

- **User to User Credentials:** One-to-One
  - Each User has one set of Credentials
  - Foreign Key: UserID in User Credentials references UserID in User
- **User to Transaction:** One-to-Many
  - Each User can have multiple Transactions (as sender or receiver)
  - Foreign Keys: SenderID and ReceiverID in Transaction reference UserID in User

NOTE: User to Transaction has two relationships (as sender and receiver).

3.6 Design Constraints

This section specifies the constraints imposed on the system design by external standards, regulatory requirements, and project limitations. These constraints must be strictly adhered to during the implementation phase.

3.6.1 Architectural Constraints

- **Centralized Data Model:** The system must utilize a single, centralized database to maintain global consistency of user balances and transaction history.
- **Server-Side Logic:** All financial calculation logic, including account balance updates and transaction verification, must be executed on the server-side.
- **Zero-Sum Principle:** Distributed ledger technology (if used) or internal accounting must adhere to a zero-sum principle where the total amount of currency in the system remains constant unless explicitly minted or burned by an authorized administrator.

3.6.2 Data Implementation Constraints

- **Fixed-Point Arithmetic:** To prevent rounding errors inherent in floating-point arithmetic, all monetary values must be stored and processed using integer-based fixed-point arithmetic. Currency conversions that result in values with more than two decimal places should be rounded to the nearest cent, following the IEEE 754 standard.
- **Atomic Transactions:** All financial operations (transfers, payments, adjustments) must be executed as atomic transactions. Either all parts of the transaction succeed, or the entire transaction is rolled back to its original state.

3.6.3 Security Constraints

- **Secure PIN Storage:** User PINs must never be stored in plain text.
- **Communication Security:** All network communication between the mobile application and the backend servers must be encrypted.
- **Authentication Throttling:** The backend must implement rate limiting and account lockout mechanisms to prevent brute-force attacks against user PINs or authentication tokens.

3.6.4 Regulatory and Standards Constraints

- **GDPR Compliance:** The system design must incorporate "Privacy by Design" principles. It must allow for the export and deletion (right to be forgotten) of personal user data upon request, where not in conflict with financial record-keeping laws.
- **PCI DSS:** Although the system may not directly handle credit card numbers if using a payment processor, the environment must adhere to PCI DSS 4.0 standards for secure coding and data protection.

3.6.5 Development Constraints

- **BDD Methodology:** Functional requirements must be specified using Behavior-Driven Development (BDD) syntax (Gherkin).
- **Test Coverage:** The codebase must maintain a minimum of 90% unit test coverage for all financial logic modules.

3.7 Software System Attributes

This section describes the non-functional quality attributes that define system excellence. These attributes have been identified through requirements analysis and prioritized using weighted criteria.

3.7.1 Justification for the Priority Matrix

The Quality Attributes Priority Matrix was developed to ensure that the most critical aspects of the system are addressed effectively. The weights assigned to each criterion reflect their relative importance:

- **Risk and Technical Complexity (Weight = 1):** This criterion evaluates the technical challenges and potential risks associated with implementing the attribute. A lower weight reflects its relative importance compared to other criteria.
- **Business Value (Weight = 3):** This measures the attribute's contribution to achieving the organization's goals and ensuring sustainability. A higher weight emphasizes its significance to the business.
- **User Impact (Weight = 5):** This assesses the attribute's importance to the end-user experience and satisfaction. The high weight highlights the priority given to user-centric design.
- **Security (Weight = 8):** This ensures the protection of data and system integrity. The highest weight underscores the criticality of security in the system.

3.7.2 Quality Attributes Priority Matrix

The following quality attributes have been prioritized using weighted scoring (based on risk, business value, user impact, and security):

Quality Attribute	**Risk (W=1)**	**Business (W=3)**	**User (W=5)**	**Security (W=8)**	**Total Score**	**Priority**
Integrity	4	5	4	5	72	1st (Highest)
Reliability	4	4	4	4	60	2nd
Usability	2	4	5	2	55	3rd
Efficiency	3	3	4	3	51	4th
Interoperability	3	3	4	2	48	5th
Maintainability	5	3	2	2	34	6th (Lower)

The total scores for each attribute were calculated by summing the weighted contributions of these criteria.

3.7.3 1st Priority: Integrity

Definition: Integrity ensures that data remains accurate, consistent, and trustworthy throughout its lifecycle. For a financial system, this means every transaction updates account balances correctly, maintains the Zero Sum principle, and prevents unauthorized modifications.

Concrete Scenario (Provided in Section 3.3): User attempts to alter a transaction after confirmation while system is in normal operation. System blocks any unauthorized modification. No transaction history alterations occur.

3.7.4 2nd Priority: Reliability

Definition: Reliability ensures the system operates consistently and correctly under specified conditions over an extended period. For a payment system, reliability means transactions never fail silently and the system maintains availability.

Concrete Scenario: User initiates multiple transactions during peak usage period. System processes all transactions correctly without data corruption or loss. All transactions complete with explicit success/failure confirmation.

3.7.5 3rd Priority: Usability

Definition: Usability measures how effectively, efficiently, and satisfactorily users can accomplish their goals. For a payment application, this means intuitive interfaces, clear feedback, and error prevention.

Concrete Scenario (Provided in Section 3.3): New user transfers high-value amount (5,000 EUR). System guides step-by-step with clear confirmations. User completes transfer successfully on first attempt with 95% success rate across all users.

3.7.6 4th Priority: Efficiency

Definition: Efficiency refers to system performance under load. For payment systems, efficiency means processing transactions quickly even during peak periods.

Concrete Scenario (Provided in Section 3.4): During peak shopping hours, 10,000 concurrent users initiate payments. System processes 500 requests per second. All payments complete within 2 seconds (p99 latency). Zero timeouts occur.

3.7.7 5th Priority: Interoperability

Definition: Interoperability ensures the system can communicate and coordinate with external systems (banks, payment terminals, etc.) using standardized interfaces.

3.7.8 6th Priority: Maintainability

Definition: Maintainability measures how easily the system can be modified, corrected, and updated. Low priority due to security and integrity constraints.

3.7.9 Identified Quality Attribute Conflicts

	Integrity	**Reliability**	**Usability**	**Efficiency**	**Interop.**	**Maintain.**
Integrity		+		−	−	−
Reliability	+		+	−	+	−
Usability		+		−	+	+
Efficiency	−	−	−		−	−
Interop.	−	+	+	−		+
Maintain.	−	−	+	−	+	

Legend: + = Supportive, − = Conflicting, blank = Neutral

Conflict Resolution Strategy:

1. **Conflict: Integrity Usability**
   - **Description:** Strict validation rules ensure integrity but may frustrate users with frequent error messages
   - **Resolution Strategy:** Integrity takes priority. However, implement error prevention through input constraints and helpful messages rather than strict validation failures
   - **Implementation:** Real-time input validation with suggestions (e.g., auto-complete for recipient names)
2. **Conflict: Usability Efficiency**
   - **Description:** User-friendly interfaces require additional processing layers, potentially reducing response time
   - **Resolution Strategy:** Prioritize Usability. Implement efficient frontend caching and progressive loading to maintain performance
   - **Implementation:** Lazy loading for transaction history, client-side caching, skeleton screens for perceived performance
3. **Conflict: Integrity Maintainability**
   - **Description:** Rigid validation rules and audit logging make system harder to modify
   - **Resolution Strategy:** Integrity takes priority. Maintainability addressed through modular design and comprehensive documentation
   - **Implementation:** Abstract validation rules into configurable modules; maintain detailed change logs
4. **Conflict: Integrity Interoperability**
   - **Description:** External system integration may compromise data integrity
   - **Resolution Strategy:** Integrity takes absolute priority. External communications use sandboxing and strong verification
   - **Implementation:** All external data validated before integration; separate database for external data with reconciliation process
5. **Conflict: Efficiency Reliability**
   - **Description:** Reliability mechanisms (redundancy, verification) may slow response time
   - **Resolution Strategy:** Balance both. Implement redundancy only for critical operations; optimize non-critical paths

    - **Implementation:** Synchronous verification for financial transactions; asynchronous notification sending

6. **Conflict: Security Usability (Account Lockout)**
    - **Description:** Security requirement for account lockout after 3 failed attempts may lock out legitimate users
    - **Resolution Strategy:** Security takes priority. Provide alternative unlock mechanisms (email verification, recovery codes)
    - **Implementation:** 24-hour automatic unlock; customer support override; email unlock link

Prioritization Summary

Priority 1 - Non-Negotiable (Security, Integrity): No compromise possible. Account security and data integrity must be protected absolutely.

Priority 2 - High Importance (Reliability, Performance): System must be dependable and fast. Trade-offs with lower priorities possible.

Priority 3 - Important (Usability): User experience matters, but never at cost of security or integrity.

Priority 4-6 - Lower Priority (Efficiency, Interoperability, Maintainability): Optimized after higher priorities are secured.

3.8 Supporting Information

3.8.1 Information Flow Model

The Information Flow Model illustrates how data moves through the Digital Wallet System and between external actors. It depicts the critical pathways for user requests, transaction processing, and communication with external banking and payment systems. The model identifies data sources, processing nodes, and destinations, ensuring that all sensitive financial data follows secure, compliant channels. This model ensures that information architecture supports both system reliability and regulatory compliance requirements.

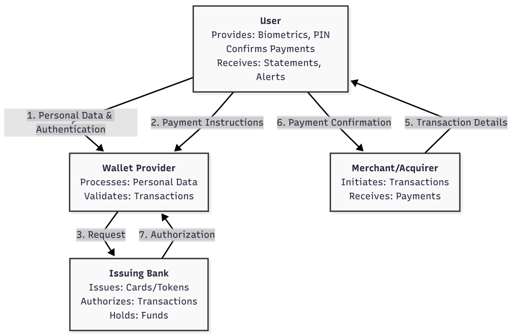

Figure 3.4: Information Flow Model

3.8.2 Cultural Model

The Cultural Model maps the external pressures acting on the system. It highlights the critical conflict between the Customer's demand for speed and the rigorous control required by Regulators and Banks. The system must resolve this tension by offering a seamless user experience that simultaneously ensures the compliance and social trust necessary for adoption.

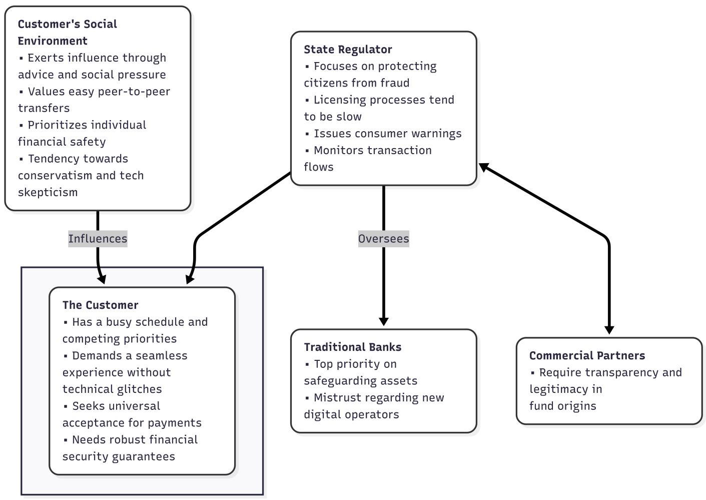

Figure 3.5: Cultural Model

Section 4: Verification (Omitted)

Note: As specified in project requirements, formal verification procedures are documented separately in BDD scenario specifications (see Section 3.2 and associated behave framework files).

Appendices

Appendix A: Assumptions and Dependencies

A.1 Assumptions

1. **User Behavior Assumptions:**
   - Users are expected to maintain the strict confidentiality of their authentication credentials (PIN) and prevent unauthorized access.
   - Users will verify payment details before confirming transactions
   - Users will report suspicious activity within 24 hours
2. **Technical Assumptions:**
   - Mobile devices have minimum 100 MB free storage
   - Network latency $\leq$ 100ms for 95% of users
3. **Organizational Assumptions:**
   - Banking partners will provide account verification within 5 seconds
4. **Legal/Compliance Assumptions:**
   - System operates within EU jurisdiction with GDPR compliance
   - PCI DSS compliance required for payment data handling

A.2 Critical Dependencies

1. **Banking System Integration:**
   - Dependency on external bank APIs for account verification
2. **Payment Terminal Support:**
   - Dependency on third-party terminals supporting QR Code and NFC
   - Mitigation: Vendor testing and certification process
3. **Mobile Platform Support:**
   - Dependency on iOS and Android platform updates
   - Mitigation: Maintain compatibility with N-2 versions of each platform
4. **Regulatory Compliance:**
   - Dependency on compliance with evolving financial regulations
   - Mitigation: Legal review process for each major release

A.3 Dependency Matrix

Component	Depends On	Criticality	Mitigation
Mobile App	iOS/Android SDKs	High	Version compatibility testing
Backend API	Database	Critical	Replication & backup
Payment Processing	Bank APIs	Critical	Circuit breaker pattern
Authentication	TLS certificates	High	Certificate rotation process
Data Storage	Cloud Infrastructure	Critical	Multi-region replication

Appendix B: Acronyms and Abbreviations

Acronym	**Definition**
ACID	Atomicity, Consistency, Isolation, Durability (database transaction properties)
AML	Anti-Money Laundering
API	Application Programming Interface
BDD	Behavior Driven Development
CSV	Comma-Separated Values
EUR	Euro (currency code)
FCTUC	Faculdade de Ciências e Tecnologia da Universidade de Coimbra
GDPR	General Data Protection Regulation (EU data protection law)
IBAN	International Bank Account Number
JSON	JavaScript Object Notation
JWT	JSON Web Token
NFC	Near Field Communication
OAuth	Open Authorization protocol
P2P	Peer-to-Peer
PDF	Portable Document Format
PCI DSS	Payment Card Industry Data Security Standard
PIN	Personal Identification Number
QR	Quick Response (code)
RBAC	Role-Based Access Control
REST	Representational State Transfer
RFP	Request for Proposals
RPO	Recovery Point Objective
RTO	Recovery Time Objective
SDK	Software Development Kit
SLA	Service Level Agreement
TLS	Transport Layer Security
TPA	Third-Party Authority / Third-Party Payment Terminal
UC	Use Case
UUID	Universally Unique Identifier
WCAG	Web Content Accessibility Guidelines